

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · DPP SHEET

# Chapter 1: A Square and A Cube

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Which of these cannot be perfect squares: 1057, 2304, 7928, 4000?
2. What is the units digit of  $87^2$  and of  $126^2$ ?
3. Given  $60^2 = 3600$ , find  $61^2$  by adding one odd number.
4. How many numbers lie between  $31^2$  and  $32^2$ ?
5. Find  $\sqrt{2025}$  by prime factorisation.
6. Find the smallest number by which 2352 must be multiplied to get a perfect square, and the square root of the product.
7. Between which two consecutive whole numbers does  $\sqrt{150}$  lie?
8. Find  $\sqrt[3]{9261}$ .
9. What is the smallest number by which 392 must be multiplied to get a perfect cube?
10. True or false: (i) a perfect cube can end in 8; (ii) every cube number has an odd number of factors.

## Answer Key

---

1. 1057, 7928 and 4000
2. 9 and 6
3.  $3600 + 121 = 3721$
4. 62
5.  $2025 = 3^4 \times 5^2$ , so the root is  $3 \times 3 \times 5 = 45$
6.  $2352 = 2^4 \times 3 \times 7^2$ ; multiply by 3; root = 84
7. 12 and 13
8.  $9261 = 3^3 \times 7^3$ , so the cube root is 21
9.  $392 = 2^3 \times 7^2$ ; multiply by 7
10. (i) True, for example  $2^3 = 8$ . (ii) False, for example 8 has 4 factors.

MathSetu